A missing spectral arc at finite cycle affinity in four-state Markov generators

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Abstract

Cycle affinities constrain the decay rates and oscillation frequencies of a Markov process, but a spectral localization bound need not describe all attainable eigenvalues. We disprove the proposed equality between the finite-affinity spectral set and a vertical compression of the unrestricted set, already for four states. At maximum allowed escape rate one and cycle budget $B_* = (\log 3)/2$, an entire sufficiently small compressed Karpelevich arc near $-1 + \mathrm{i}/2$ is absent. The exclusion holds in a compact class defined by all oriented cycle-product inequalities, including reducible and zero-rate limits. The proof combines the equality case of cycle-winding localization with a uniform quadratic expansion about a biased square. Opening chords creates triangles whose affinity constraints give an upper quadratic coefficient $1/5$, whereas the compressed arc has coefficient $1/2$. A coercive estimate controls unequal escape rates, unused affinity, and arbitrary approaches to the endpoint. The validity interval is existential: no numerical radius, sharpness of $1/5$, or full boundary classification is asserted.

Keywords: Markov generators; cycle affinity; spectral localization; nonequilibrium relaxation; inverse spectral problem.

1 Spectral localization and attainability

An eigenvalue of a continuous-time Markov generator specifies a possible decay rate and, when it is nonreal, an oscillation frequency. Thermodynamic restrictions on these quantities provide information about relaxation without specifying all transition rates. Uhl and Seifert proposed an affinity-dependent spectral ellipse and studied perturbations of a uniform biased cycle [1]. Xu *et al.* established an ellipse based on edge forces and posed a stronger question about the complete spectral set: does vertical compression of the unrestricted region give exactly the finite-force region [2, End Matter and Supplementary Eq. (S3)]? The later cycle-winding localization theorem of Kolchinsky, Ohga and Ito proves the cycle-affinity ellipse for general finite generators [3, Theorem 3 and Table I]. An outer localization theorem, however, does not establish attainability of every point in a proposed region.

Here we address this distinction for four states. We exhibit a continuum of points in the proposed compressed region that no admissible generator can realize. These points approach the spectrum of a uniform biased square. At that endpoint the known localization theorem is rigid. Nearby, the triangle affinities created by adding chords restrict the quadratic spectral response. Controlling these new constraints over all supports, rather than assuming a fixed ring, is essential to the exclusion.

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1.1 Conventions and the spectral-set question

Use a row generator $Q = (q_{ij})_{i,j=0}^{3}$: for $i \neq j$, $q_{ij} \geq 0$ is the rate from i to j, and $q_{ii} = -\sum_{j\neq i} q_{ij}$. Column probabilities obey $\dot{p} = Q^{\mathsf{T}} p$. Transposition preserves the spectrum, and backward observables satisfy $\dot{f} = Qf$. The maximum escape rate is

$$r(Q) = \max_i \sum_{j\neq i} q_{ij}. \tag{1}$$

Rates and eigenvalues have inverse-time units. We normalize by the allowed maximum escape rate, set that bound to one, and henceforth use dimensionless rates and eigenvalues. Physical units can be restored by multiplying every rate and eigenvalue by the same positive rate scale.

For an irreducible generator with paired support ($q_{ij} > 0$ if and only if $q_{ji} > 0$), define, for every present simple oriented cycle γ of length $\ell_\gamma \geq 3$,

$$F_\gamma = \log \frac{\prod_{(i\to j)\in\gamma} q_{ij}}{\prod_{(i\to j)\in\gamma} q_{ji}}, \qquad B(Q) = \max_\gamma \frac{|F_\gamma|}{2\ell_\gamma}. \tag{2}$$

Set $B(Q) = 0$ when no such cycles occur. On four states their lengths are three or four; backtracking two-cycles have zero affinity. Both orientations are included. Stationary-probability factors telescope around a cycle, so under local detailed balance F_γ is the entropy change in the environment per completed cycle, in units of k_{B}. The spectral argument uses the rate ratios and does not require an additional reservoir model.

Let $\Sigma_{4,B}$ be the union of individual eigenvalues over this irreducible, paired-support class with $r(Q) = 1$ and $B(Q) = B$. Let Ω_4' be the corresponding union over all four-state generators with $r(Q) = 1$, without an affinity restriction and allowing irreversible or reducible matrices. Define

$$D_B(x + \mathrm{i}y) = x + \mathrm{i}\tanh(B)y. \tag{3}$$

The equality under consideration is

$$\Sigma_{4,B} \stackrel{?}{=} D_B(\Omega_4') \qquad \text{for every finite } B \geq 0. \tag{4}$$

This is a question about individual eigenvalues, not prescribed complete spectra. We keep exact maximum rate and exact affinity on the left of (4). The exclusion below holds in a larger bounded class, so it also covers these exact constraints.

The unrestricted spectral region is related to the stochastic Karpelevich region by uniformization. We need only one explicit zero-diagonal stochastic companion to prove membership in Ω_4'. We therefore do not assume a global equivalence between the exact-rate and bounded-rate spectral unions.

2 The missing arc

Fix $B_* = (\log 3)/2$, for which $\tanh B_* = 1/2$. Define $\mathcal{K}$ as the set of four-state row generators with $r(Q) \leq 1$ and the polynomial inequalities

$$\prod_{(i\to j)\in\gamma} q_{ij} \leq 3^{\ell_\gamma} \prod_{(i\to j)\in\gamma} q_{ji} \quad \text{for every oriented simple three- or four-cycle } \gamma. \tag{5}$$

These conditions are imposed even when a product vanishes. The set $\mathcal{K}$ is a closed subset of the bounded rate cube, hence compact. It contains the closure of the paired-support class with $r \leq 1$ and $B \leq B_*$; we do not need or claim equality with that closure. One-way edges between components are allowed, but no positive directed cycle can contain an irreversible edge: its positive forward product in (5) forces every reverse rate positive.

Theorem 1. *There exists $\eta \in (0, 1/2)$ such that every $Q \in \mathcal{K}$ and every $\lambda \in \operatorname{spec}(Q)$ with $\delta = -1 - \operatorname{Re}\lambda \in (0, \eta)$ obey*

$$|\operatorname{Im}\lambda| \le \frac{1}{2} - \frac{\delta}{2} + \frac{\delta^2}{4}. \tag{6}$$

For every $\delta \in (0, \eta)$, the point

$$\Lambda(\delta) = -1 - \delta + \frac{\mathrm{i}}{2}\sqrt{1 - 2\delta + 3\delta^2} \tag{7}$$

belongs to $D_{B_*}(\Omega_4')$ and lies outside $\bigcup_{Q\in\mathcal{K}} \operatorname{spec}(Q)$. Each such point has positive distance from that spectral union. In particular, (4) is false.

The proof gives the stronger uniform asymptotic estimate

$$h \le \frac{\delta^2}{5} + C_2\delta^3, \qquad h = \operatorname{Im}\lambda - \frac{1}{2} + \frac{\delta}{2}, \tag{8}$$

for $h \ge 0$ and sufficiently small positive δ, with one finite constant $C_2 \ge 0$ over the full class. By comparison,

$$\operatorname{Im}\Lambda(\delta) = \frac{1}{2} - \frac{\delta}{2} + \frac{\delta^2}{2} + O(\delta^3). \tag{9}$$

The coefficient $1/5$ is an upper bound; its attainability is not proved. Neither η nor the positive distance in the theorem is supplied numerically. The argument does not decide an arbitrary fixed interior point of the arc.

3 Rigidity of the limiting generator

We use the following published localization input [3, Theorem 3]. For a nonreal eigenvalue $x + \mathrm{i}y$, $y > 0$, of a finite generator with maximum escape rate r, and each $\tau \in [0, 1]$, a certifying cycle γ of length ℓ and eigenvector winding $w \in \{1, \dots, \lceil \ell/2 \rceil - 1\}$ satisfies

$$\tau\frac{y}{-x}\tan\frac{\pi w}{\ell} + (1-\tau)\frac{y}{2r+x}\cot\frac{\pi w}{\ell} \le \tanh\frac{F_\gamma}{2\ell}. \tag{10}$$

The theorem applies to arbitrary finite generators, and its left-eigenvector convention for Q^{T} is the right-eigenvector equation $Qf = \lambda f$ used here. For $Q \in \mathcal{K}$ every positive cycle has finite affinity by (5).

For a nonreal eigenvalue, $-2r < x < 0$. Choose $\tau = -x/(2r)$ in (10). Its left side becomes $y/[r\sin(2\pi w/\ell)]$. Thus

$$y \le r/2 \le 1/2 \qquad (Q \in \mathcal{K}). \tag{11}$$

If $\lambda_0 = -1 + \mathrm{i}/2$ occurs, necessarily $r = 1$. With $\tau = 1/2$, the left side of (10) for a three-cycle is $1/\sqrt{3} > 1/2$. A certifying cycle must therefore have length four, winding one, and affinity $4\log 3$.

Lemma 1. *Every generator in $\mathcal{K}$ with eigenvalue $\lambda_0 = -1 + \mathrm{i}/2$ is, up to a permutation of the states, the uniform biased square*

$$Q_0 = \begin{pmatrix} -1 & 3/4 & 0 & 1/4 \\ 1/4 & -1 & 3/4 & 0 \\ 0 & 1/4 & -1 & 3/4 \\ 3/4 & 0 & 1/4 & -1 \end{pmatrix}. \tag{12}$$

Proof. We retain the equality information in the finite-affinity proof of (10) [3, SM3, Eq. (S55), and SM5]. On the certifying Hamiltonian cycle, write the eigenvector ratios as $f_{j+1}/f_j = \exp(-\xi_j + \mathrm{i}\phi_j)$. At the present endpoint the source parameters are $\gamma = 0$ and $\sigma = 1/2$, and its cycle construction yields

$$4\log 3 \geq F_\gamma \geq \sum_{j=0}^{3} J(\phi_j, \xi_j) \geq 4J(\pi/2, 0) = 4\log 3, \tag{13}$$

where

$$J(\phi, \xi) = \log \frac{e^{\xi}\sin\phi + 1/2}{e^{-\xi}\sin\phi - 1/2}, \qquad 0 < \phi < \pi, \quad \xi < \log(2\sin\phi). \tag{14}$$

The phase and log-modulus averages are $(\pi/2, 0)$. The open domain is convex, and J is strictly convex there; an explicit Hessian check is given in Appendix A. Equality in (13) forces $(\phi_j, \xi_j) = (\pi/2, 0)$ for all four edges. Thus f is a nonzero multiple of $(1, \mathrm{i}, -1, -\mathrm{i})^{\mathsf{T}}$ in cycle order.

Let $a_j = q_{j,j+1}$, $b_j = q_{j,j-1}$ and $c_j = q_{j,j+2}$, with indices modulo four. The certifying ring edges and their reverses are positive. Dividing the jth eigenvector equation by f_j gives

$$a_j + b_j + 2c_j = 1, \qquad a_j - b_j = 1/2. \tag{15}$$

Hence $a_j = 3/4 - c_j$, $b_j = 1/4 - c_j > 0$, and

$$\frac{a_j}{b_j} = \frac{3/4 - c_j}{1/4 - c_j} \geq 3, \tag{16}$$

with strict inequality for $c_j > 0$. The Hamiltonian cycle has $\prod_j a_j/b_j \leq 3^4$. Every chord rate therefore vanishes, giving (12). This also excludes other reducible or zero-rate limits at the endpoint. □

The characteristic polynomial of Q_0 is $\lambda(\lambda+2)((\lambda+1)^2 + 1/4)$. Its spectrum reproduces the uniform-cycle formula in Ref. [3, Eq. (S66)]; in particular, λ_0 is simple. Compactness now implies that any sequence of generators in $\mathcal{K}$ with eigenvalues tending to λ_0 has distance tending to zero from the finite set of labelled copies of Q_0.

4 Uniform expansion with escape and affinity slack

In a fixed neighborhood of Q_0 all eight ring rates remain positive. Let $\overline{x} = \frac{1}{4}\sum_{j=0}^{3} x_j$, and introduce

$$\begin{aligned} s_j &= 1 - \sum_{k \neq j} q_{jk} \geq 0, & v_j &= \tfrac{1}{2}\log(a_j/b_j) - B_*, & \\ m &= -\overline{v} \geq 0, & w_j &= v_j + m, & \overline{w} &= 0. \end{aligned} \tag{17}$$

Here $m \geq 0$ follows exactly from the Hamiltonian cycle inequality. The four nonnegative c_j describe both directions of both chords. All nearby generators are represented by

$$\begin{aligned} a_j &= \frac{1 - s_j - c_j}{2}\left[1 + \tanh(B_* + w_j - m)\right], \\ b_j &= \frac{1 - s_j - c_j}{2}\left[1 - \tanh(B_* + w_j - m)\right]. \end{aligned} \tag{18}$$

Set $a = 3/4$, $b = 1/4$, and $u_j = 2ab\, w_j = 3w_j/8$, so $\overline{u} = 0$. With fixed Euclidean norms, write

$$\mathcal{N} = \|c\| + \|s\| + \|u\| + m. \tag{19}$$

All remainder constants below are uniform in a sufficiently small fixed neighborhood. Indeed, (18) extends analytically to signed coordinates in an open neighborhood. Simplicity of λ_0 gives one analytic upper eigenvalue branch there, with bounded third derivatives on a smaller closed ball. Feasibility is imposed after this analytic extension; no differentiable parametrization of feasible matrices is assumed.

The homogeneous linear coefficient of that eigenvalue is

$$\lambda^{(1)} = -(1 + \mathrm{i}/2)\overline{c} + (1 - \mathrm{i}/2)\overline{s} - \frac{3\mathrm{i}}{4}m. \tag{20}$$

Consequently, for $\delta = -1 - \operatorname{Re}\lambda$ and $h = \operatorname{Im}\lambda - 1/2 + \delta/2$,

$$\delta = \overline{c} - \overline{s} + O(\mathcal{N}^2), \tag{21}$$

$$h = -\overline{s} - \frac{3}{4}m + K(c,u) + O((\overline{s}+m)\mathcal{N} + \mathcal{N}^3). \tag{22}$$

Every quadratic term omitted from K contains an escape or affinity slack. Nonnegativity of s_j and m bounds their sum by a constant times $(\overline{s}+m)\mathcal{N}$. Thus (22) retains unequal escape rates and unused affinity explicitly.

4.1 The quadratic form

The following formulas specify the complete slack-free quadratic form. Let V have zero diagonal and entries

$$V_{j,j+1} = -ac_j + u_j, \qquad V_{j,j-1} = -bc_j - u_j, \qquad V_{j,j+2} = c_j. \tag{23}$$

Let $U_{jk} = \mathrm{i}^{jk}/2$, $M = U^*VU$, and $(\ell_0, \ell_1, \ell_2, \ell_3) = (0, -1 + \mathrm{i}/2, -2, -1 - \mathrm{i}/2)$. The standard simple-root expansion in this diagonal basis gives

$$W = \sum_{k\neq 1} \frac{M_{1k}M_{k1}}{\ell_1 - \ell_k},$$

$$K(c,u) = \operatorname{Im} W - \tfrac{1}{2}\operatorname{Re} W - \frac{1}{4}\sum_{j=0}^{3}\left(2c_ju_j + \frac{8}{3}u_j^2\right). \tag{24}$$

The last term comes from the nonlinear coordinates, not from the resolvent: $\tanh(B_* + v) = 1/2 + 3v/4 - 3v^2/8 + O(v^3)$, and its product with $1 - s_j - c_j$ gives the intrinsic quadratic rate correction. Appendix B supplies a second representation of (24) using characteristic invariants.

Eliminate $u_3 = -u_0 - u_1 - u_2$ and write $\widehat{u} = (u_0, u_1, u_2)^{\mathsf{T}}$. Direct expansion of (24) gives the Hessian

$$H = \begin{pmatrix} -14/3 & -4/3 & -10/3 \\ -4/3 & -8/3 & -4/3 \\ -10/3 & -4/3 & -14/3 \end{pmatrix}. \tag{25}$$

Its leading principal minors are $-14/3$, $32/3$, and $-640/27$; therefore it is negative definite. The unique maximizer at fixed c is

$$\begin{aligned} u_0^* &= (-3c_0 + 6c_1 + 3c_2 - 6c_3)/40, \\ u_1^* &= (-6c_0 - 3c_1 + 6c_2 + 3c_3)/40, \\ u_2^* &= -u_0^*, \qquad u_3^* = -u_1^*. \end{aligned} \tag{26}$$

Define opposite-pair sums and differences by

$$S = c_0 + c_2, \quad T = c_1 + c_3, \quad D = (c_0 - c_2)/2, \quad E = (c_1 - c_3)/2. \tag{27}$$

Exact completion of the square yields

$$K(c,u) = K_{\max}(c) + \tfrac{1}{2}(\widehat{u} - \widehat{u}^*)^{\mathsf{T}} H (\widehat{u} - \widehat{u}^*), \tag{28}$$

$$K_{\max}(c) = \frac{2}{5}(D^2 + E^2) - \frac{5}{64}(S - T)^2. \tag{29}$$

These are homogeneous polynomial identities, with no constraint on $\sum_j c_j$. In particular, the optimization does not impose the other cycle inequalities or assert that its maximizing direction is realized by an exact feasible family.

4.2 A coercive bound fixes the scale of approach

A quadratic expansion alone would not constrain arbitrary approaches to the endpoint: initially a rate deviation could be much larger than δ. We now control that possibility. Put

$$C = \overline{c}, \qquad L = \overline{s} + m, \qquad U_0 = \|u\|. \tag{30}$$

Since $c_j \geq 0$, $\|c\| \leq 4C$. Negative definiteness in (28), linearity of $u^*(c)$, and Young's inequality give constants $A, k > 0$ such that

$$K(c,u) \leq AC^2 - kU_0^2. \tag{31}$$

Norm equivalence and nonnegative slacks also give $\mathcal{N} \leq C_0(C + L + U_0)$. In a ball $\mathcal{N} \leq \varepsilon$,

$$L\mathcal{N} \leq \varepsilon L, \qquad \mathcal{N}^3 \leq C_3\varepsilon(C^2 + U_0^2) + C_4\varepsilon^2 L. \tag{32}$$

For example, cube the preceding norm bound, absorb mixed terms into the individual cubes, and use $C, U_0, L = O(\varepsilon)$. Choosing ε sufficiently small in (22)–(32) absorbs the slack and U_0^2 remainders into the negative terms. For fixed $A_1, k_1 > 0$,

$$h \leq A_1 C^2 - k_1(L + U_0^2). \tag{33}$$

On the part with $h \geq 0$, this implies $L = O(C^2)$, $U_0 = O(C)$, and $\mathcal{N} = O(C)$. Equation (21) then gives $\delta = C + O(C^2)$. If $C = 0$, (33) forces every coordinate to vanish and hence $\delta = 0$. If $C > 0$ is sufficiently small, then $C/2 \leq \delta \leq 3C/2$. Substitution proves the uniform estimates

$$C = \delta + O(\delta^2), \qquad \mathcal{N} = O(\delta), \qquad L = O(\delta^2). \tag{34}$$

No smoothness of a sequence of feasible generators enters this argument.

5 Triangle constraints and exclusion

5.1 The triangles opened by chords

The cycle $j \to j+1 \to j+2 \to j$ in (5) gives

$$a_j a_{j+1} c_{j+2} \leq 27 b_{j+1} b_{j+2} c_j. \tag{35}$$

The forward triangle starting at $j+2$ supplies the opposite comparison. Since $a_j = 3/4 + O(\mathcal{N})$ and $b_j = 1/4 + O(\mathcal{N})$, and ring rates stay bounded away from zero, these inequalities imply

$$c_{j+2} \leq 3[1 + O(\mathcal{N})]c_j, \qquad c_j \leq 3[1 + O(\mathcal{N})]c_{j+2}. \tag{36}$$

Only positive ring rates were divided out. Thus the estimates remain valid on zero-chord faces; one zero direction forces its reverse to vanish. In the variables (27),

$$|D| \leq S/4 + O(\mathcal{N})S, \qquad |E| \leq T/4 + O(\mathcal{N})T. \tag{37}$$

Combining this with the nonpositive square in (28) gives

$$K(c,u) \le \frac{(S+T)^2}{80} - \frac{21}{320}(S-T)^2 + O(\mathcal{N})C^2$$
$$\le \frac{C^2}{5} + O(\mathcal{N})C^2, \tag{38}$$

because $S+T=4C$. The other cycle constraints can only decrease the allowed class.

On $h \ge 0$, apply (34) to (22) and (38). The negative linear slack terms may be dropped, while all remaining errors are $O(\delta^3)$. This proves

$$h \le \frac{\delta^2}{5} + O(\delta^3) \tag{39}$$

uniformly near every labelled copy of Q_0.

5.2 Compactness covers the full class

Suppose generators in $\mathcal{K}$ with $\delta \to 0^+$ and $h \ge 0$ could remain outside fixed small neighborhoods of those finitely many copies of Q_0. By (11), $1/2 - \delta/2 \le \operatorname{Im}\lambda \le 1/2$, so the eigenvalues tend to λ_0. A compact subsequence of generators converges, and the limit has eigenvalue λ_0 by continuity of the characteristic polynomial. Lemma 1 puts that limit in the finite set, a contradiction.

Thus one uniform constant C_2 in (8) covers all such generators for sufficiently small positive δ. Shrink $\eta < 1/2$ further so that $C_2\eta \le 1/20$. Then (6) holds for upper eigenvalues with $h \ge 0$. If $h < 0$, it is immediate. Real eigenvalues satisfy the bound as well, and negative imaginary parts follow by conjugation. This proves the exclusion bound for the entire compact class, including all support changes and reducible limits.

5.3 Exact membership of the compressed comparison arc

For $0 \le t \le 1$ consider the column-stochastic companion

$$P(t) = \begin{pmatrix} 0 & 0 & 0 & 1-t \\ 1 & 0 & 0 & t \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{40}$$

Its diagonal vanishes, so $P(t) - I$ is a column generator with every escape rate exactly one. The row generator $P(t)^{\mathsf{T}} - I$ therefore belongs to the unrestricted class defining Ω_4'. For $0 < t < 1$ it is irreducible. Its characteristic polynomial before the shift is

$$z^4 - tz - (1-t) = (z-1)(z^3 + z^2 + z + 1 - t). \tag{41}$$

This is the four-state arc polynomial obtained from the Karpelevich parametrization in Ref. [2, Supplementary Eq. (S2)] for denominators three and four. The explicit matrix proves the needed membership without appealing to a global description of Ω_4'.

For $0 < \delta < 1/2$, set

$$t(\delta) = 4\delta - 8\delta^2 + 8\delta^3, \qquad z(\delta) = -\delta + \mathrm{i}\sqrt{1 - 2\delta + 3\delta^2}. \tag{42}$$

Direct substitution into the cubic in (41) gives zero. Moreover, $t'(\delta) = 4(6\delta^2 - 4\delta + 1) > 0$, with $t(0) = 0$ and $t(1/2) = 1$. This traces the upper arc from i to $e^{2\pi\mathrm{i}/3}$. In particular, $z(\delta) - 1 \in \Omega_4'$ and $D_{B_*}(z(\delta) - 1) = \Lambda(\delta)$.

To compare with (6), both quantities to be squared are positive on this interval, and

$$(1 - 2\delta + 3\delta^2) - (1 - \delta + \delta^2/2)^2 = \delta^2 + \delta^3 - \delta^4/4 > 0. \tag{43}$$

Therefore $\operatorname{Im} \Lambda(\delta) > 1/2 - \delta/2 + \delta^2/4$. For $0 < \delta < \eta$ these points are absent from $\operatorname{spec}(\mathcal{K})$, where $\operatorname{spec}(\mathcal{K})$ denotes the union of individual spectra. Finally,

$$\{(Q, \lambda) : Q \in \mathcal{K},\ |\lambda| \le 2, \det(\lambda I - Q) = 0\} \tag{44}$$

is compact and projects onto $\operatorname{spec}(\mathcal{K})$; the modulus bound follows from Gershgorin's theorem. Each excluded point has positive distance from this compact union. Since the exact-rate, exact-affinity class is contained in $\mathcal{K}$, Theorem 1 follows.

6 Physical meaning and scope

At the endpoint, the uniform square has clockwise and counterclockwise rates $3/4$ and $1/4$. Increasing the normalized decay displacement $\delta = -1 - \operatorname{Re} \lambda$ while staying above the endpoint tangent requires chord rates of order δ. Those transitions also generate three-state cycles. Their affinity budgets restrict opposite chord rates to a ratio approaching the interval $[1/3, 3]$, which limits the second-order frequency response. This explains why a deformation based only on compressing the unrestricted shape overestimates attainability.

The distinction from earlier results is specific. The perturbation class in Uhl and Seifert [1, Sec. 4.1, condition (i), and Sec. 4.2] retains the existing edges of a ring. It does not incorporate the triangles opened here. The more recent kinetic refinement of Xu and Meng [4, Secs. II–III and Corollary 3] assumes a unicyclic generator and incorporates geometric means of forward and backward rates. Uniform-cycle rigidity within that model is not the contribution claimed here. The general localization theorem [3] is our starting point; the additional all-support curvature estimate gives the nonattainment conclusion. These comparisons concern the stated results and assumptions of the cited works, without claiming an exhaustive classification of related spectral inequalities.

The theorem concerns an exact, time-independent four-state Markov model. Second-order perturbation theory is part of a proof with a uniform remainder, not a finite-order replacement of the dynamics. It excludes normalized decay–frequency pairs even from the closure of the bounded-affinity class. It does not give the full attainable boundary, establish $1/5$ as its sharp curvature, or provide an effective value of η. In particular, the point at the companion parameter $t = 1/2$ is undecided by this proof. No general inclusion in the compressed Karpelevich region is proved or disproved: the set equality fails by nonattainment. Determining a quantitative validity interval or the actual sharp local boundary remains a separate problem.

Acknowledgements and generative AI use

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Data and code availability

The accompanying reproducibility material contains the exact symbolic and rational-arithmetic checks, their expected outputs, and instructions for regeneration. No experimental data or numerical parameter scan is used in the proof. The scripts check algebraic identities; the uniform remainder, scale control, and compactness arguments are given in the text.

A Strictness of the endpoint Jensen step

For completeness, the strict convexity needed in Lemma 1 can be checked explicitly from (14). Put $v = \sin\phi$, $\rho = e^{\xi}$, $N = \rho v + 1/2$, and $M = v/\rho - 1/2$. On the open domain in (14), $0 < v \le 1$, $0 < \rho < 2v$, and $N, M > 0$. Direct differentiation gives

$$\partial_\xi^2 J = \frac{\rho v}{2N^2} + \frac{v}{2\rho M^2} > 0 \tag{45}$$

and the specialization of the Hessian identity of Ref. [3, SM5],

$$\det \nabla^2 J = \frac{\frac{5}{4}v^2\big((\rho+\rho^{-1})/2\big)^2 + \frac{1}{4}(1-v^2)\big((\rho-\rho^{-1})/2 - 2v\big)^2}{N^3 M^3} > 0. \tag{46}$$

The first numerator term is strictly positive everywhere in the domain. The Hessian is positive definite, and the domain is convex because $\log(2\sin\phi)$ is concave on $(0,\pi)$. Hence equality in Jensen's inequality forces identical phase and log-modulus pairs.

B Reconstruction from characteristic invariants

The quadratic form may be verified without an eigenbasis. With $s = m = 0$, write $A = Q + I$. Its diagonal is zero and its row sums are one. Let

$$P_2 = \sum_{i<j} A_{ij}A_{ji}, \qquad P_3 = \sum_{i<j<k} (A_{ij}A_{jk}A_{ki} + A_{ik}A_{kj}A_{ji}). \tag{47}$$

The trace identities give the z^2 and z coefficients of $\det(zI - A)$ as $-P_2$ and $-P_3$. Since one is an eigenvalue, factoring $z - 1$ leaves

$$f(z) = z^3 + z^2 + (1 - P_2)z + 1 - P_2 - P_3. \tag{48}$$

Introduce a formal scalar ϵ by replacing (c, u) with $(\epsilon c, \epsilon u)$ in (18). Through degree two,

$$\begin{aligned} A_{j,j+1} &= \tfrac{3}{4} + \epsilon(u_j - 3c_j/4) - \epsilon^2(c_j u_j + 4u_j^2/3) + O(\epsilon^3), \\ A_{j,j-1} &= \tfrac{1}{4} + \epsilon(-u_j - c_j/4) + \epsilon^2(c_j u_j + 4u_j^2/3) + O(\epsilon^3), \\ A_{j,j+2} &= \epsilon c_j. \end{aligned} \tag{49}$$

Write $f = f_0 + \epsilon f_1 + \epsilon^2 f_2 + O(\epsilon^3)$ and $z = z_0 + \epsilon z_1 + \epsilon^2 z_2 + O(\epsilon^3)$, where $z_0 = \mathrm{i}/2$. Since $f_0'(z_0) \neq 0$,

$$z_1 = -\frac{f_1(z_0)}{f_0'(z_0)}, \qquad z_2 = -\frac{f_2(z_0) + f_1'(z_0)z_1 + \frac{1}{2}f_0''(z_0)z_1^2}{f_0'(z_0)}. \tag{50}$$

Substitution gives $z_1 = -(1 + \mathrm{i}/2)\overline{c}$ and $K(c, u) = \operatorname{Im} z_2 - \operatorname{Re} z_2/2$, reproducing (25)–(29) as identities in all seven independent variables $(c_0, c_1, c_2, c_3, u_0, u_1, u_2)$.

The supplied symbolic program implements these invariants. A separate program uses Gaussian rational arithmetic and a bordered linear system for the eigenvector derivative.

Because the resulting coefficient is a homogeneous quadratic polynomial in seven variables, the seven basis directions and their twenty-one pair sums determine all twenty-eight coefficients exactly. Five further first-order directions check the four escape slacks and the mean-affinity slack in (20). This is a complete check of the finite quadratic identity, not a fit or a numerical argument for the universal validity interval.

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